Spec → Code: ONC_ODB_010_JR_STD

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1. Spec

REPORT ID

ONC_ODB_010_JR_STD

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report ("Non-study related oncologic treatment reconciliation") identifies Oncologic Drugs and Biologics (ODB) administered within a 14-day window before or after the onset of a Serious Adverse Event (SAE) or Event of Clinical Interest (ECI), where no matching record exists in the Concomitant Medication (CM) form. It is used to flag potential under-reporting of oncologic treatments in CM data relative to ODB source records, in the context of AE/SAE follow-up.

COLUMN HEADING

Country | Site | Subject Number | ODB Visit | ODB Form | ODB Sequence | ODB Name | ODB Start Date | ODB Stop Date | AE Sequence | AE Term | AE Onset Date and Time

LOGIC

Source data:
- Subject/site reference data (subject ID, country, site number, subject number, screening number) from a subject cross-reference view.
- Visit-level date-of-visit (DOV) data, used to exclude erroneous visits (records where the DOV error flag is set are excluded).
- Adverse Event (AE) form data, used to derive SAE/ECI records: an AE is treated as an SAE if (a) its "Serious Criteria" composite flag is non-null AND its "Is this a serious AE" flag is not 'None', OR (b) it is flagged as an Event of Clinical Interest ('Yes'). Only AE records with a non-null onset date and not marked as deleted (form or add-on item) are considered; only the date portion of onset date/time is used (time disregarded).
- ODB Form (and its supplemental/repeating add-on entry form) data: includes multiple physical eCRF form variants (ODB, ODBF, ODBCF, ODBCS, ODBH, ODBHF, ODBHN, ODBNF), unified into one logical ODB dataset via a study-specific data dictionary/alias mapping (driven by external supporting documents). Key ODB fields: Standardized Oncologic Drug or Biologic Name, Imputed Start Date of Medication, Imputed Stop Date of Medication (falling back to Imputed Start/Stop Date of Regimen for older studies where medication-level dates aren't available). Deleted forms/items and erroneous visits (per DOV error flag) are excluded.
- Concomitant Medication (CM) form data: Medication Category, Standardized Medication Name, Imputed Start Date. Deleted/erroneous records excluded similarly.

Matching/filtering logic:
1. For each ODB drug/biologic record (with non-null standardized name and non-null computed start date, where start date = coalesce(medication start date, regimen start date), and stop date = coalesce(medication stop date, regimen stop date), defaulting stop date to current system date if still null), join to AE records for the same subject where the AE qualifies as an SAE/ECI (per definition above) and has a non-null onset date.
2. Keep only combinations where the ODB drug's date range (start date to stop date) overlaps the SAE/ECI window (SAE onset date minus 14 days to SAE onset date plus 14 days) -- i.e., exclude when ODB stop date is before (onset-14) or ODB start date is after (onset+14).
3. Left-join to CM records for the same subject where CM Medication Category = 'Oncologic' (case-insensitive), CM Standardized Medication Name equals the ODB drug's standardized name (case-insensitive), and CM Imputed Start Date equals the ODB record's computed start date (date only, time disregarded).
4. Exclude any ODB/SAE combination where a matching CM record was found (i.e., only keep rows with no matching CM record) -- this identifies oncologic drugs given around the SAE/ECI window that are NOT reconciled in the CM form.
5. Optionally restrict to ODB records from a specific list of ODB form mnemonics if the %odbform macro variable is supplied; otherwise all ODB form variants are included.
6. Join to subject reference data to bring in country/site/subject number, restricting to subjects present in the subject cross-reference dataset (inner-style merge with "if a and b").

7. Apply a study-configurable subdataset filter (%subdataset macro) as an additional row restriction.

Derived/display fields: ODB Visit (visit mnemonic, with 'CCRF' relabeled as 'Common'), ODB Sequence displayed as "formidx(itemsetidx)", ODB Stop Date displayed as 'Continuing' when the underlying stop date was originally null (i.e., defaulted to system date).

REPORT SCENARIOS

1. No qualifying records found (0 observations after all filtering): output a single-message placeholder report/sheet stating "No records meet the test condition" instead of the detail listing.
2. Qualifying records exist: output the full detail listing of ODB drug/biologic administrations near an SAE/ECI window with no corresponding CM record, one row per Subject/ODB record/AE combination.
3. Optional form-filtering scenario: if a specific ODB form list is supplied via parameter, restrict to those ODB form mnemonics; otherwise report covers all ODB form variants (displayed as "All" in the title).
4. Older-study scenario: when medication-level imputed start/stop dates are not captured on the ODB form, the report falls back to using the regimen-level imputed start/stop dates instead.
5. Missing ODB stop date scenario: when ODB stop date is null, it is treated as ongoing and defaults to the current system date for interval-overlap comparison, but is displayed to the user as "Continuing" rather than a date.

SORTING

Intermediate result (finalini) sorted by subjectid. Final output sorted by Country, Site Number, Subject Number, ODB Visit, ODB Sequence (form-level id), ODB Add-on Sequence (item-set id), AE Sequence, ODB Form.

2. Logic Understanding

This report cross-references the ODB (Oncologic Drugs and Biologics) form against the AE form to find oncologic drug administrations occurring within a 14-day window of a qualifying SAE/ECI onset, then checks the CM (Concomitant Medication) form via a left-join on category/name/start-date for a matching reconciliation record. Any ODB/SAE-ECI combination lacking a matching CM record is flagged as a potential under-reporting gap and included in the output, with data-quality exclusions (deleted records, erroneous visits) and optional macro-driven filters (form list, subdataset) applied along the way, plus fallback logic for older studies and 'Continuing' display for open-ended medications. If no records survive filtering, a placeholder message is shown instead of a detail listing.
1. Start by assembling four source datasets: subject/site cross-reference data (subject ID, country, site, subject number, screening number), visit-level DOV data (to identify erroneous visits), AE form data, and ODB form data (unified across its 8 physical eCRF variants: ODB, ODBF, ODBCF, ODBCS, ODBH, ODBHF, ODBHN, ODBNF) plus CM form data.

2. From the AE data, derive which AE records qualify as SAE/ECI: an AE counts if (a) its Serious Criteria composite flag is non-null AND its 'Is this a serious AE' flag is not 'None', OR (b) its Event of Clinical Interest flag = 'Yes'. Only keep AE records that also have a non-null onset date and are not marked deleted (form-level or add-on item-level); disregard the time portion of onset date/time, using only the date.

3. From the ODB data, exclude deleted forms/items and records tied to visits flagged as erroneous per the DOV error flag. For each remaining ODB record compute: start date = coalesce(Imputed Start Date of Medication, Imputed Start Date of Regimen), stop date = coalesce(Imputed Stop Date of Medication, Imputed Stop Date of Regimen); if stop date is still null after that, default it to the current system date (representing an ongoing medication). Only keep ODB records with a non-null standardized name and non-null computed start date.

4. Join each qualifying ODB record to AE/SAE-ECI records for the same subject (from step 2), requiring the AE to have a non-null onset date.

5. Filter these ODB/AE combinations to keep only those where the ODB drug's date range (computed start date to computed stop date) overlaps the SAE/ECI window of onset date minus 14 days to onset date plus 14 days -- concretely, exclude the pair if the ODB stop date falls before (onset - 14) or the ODB start date falls after (onset + 14).

6. For the surviving ODB/AE combinations, left-join to CM records for the same subject where CM Medication Category = 'Oncologic' (case-insensitive), CM Standardized Medication Name equals the ODB record's standardized name (case-insensitive), and CM Imputed Start Date equals the ODB record's computed start date (comparing dates only, ignoring time).

7. Exclude any ODB/SAE combination where a matching CM record was found via the left-join in step 6 -- only rows with NO matching CM record are retained, since these represent oncologic drugs given around the SAE/ECI window that are not reflected/reconciled in CM.

8. If the %odbform macro variable is supplied, further restrict the surviving records to only those ODB records whose form mnemonic is in the supplied list; otherwise keep all ODB form variants (and the report title will display 'All').

9. Merge the surviving records with the subject cross-reference dataset to bring in Country, Site, and Subject Number, using an inner-style merge ('if a and b') so that only subjects present in the cross-reference data are retained.

10. Apply the study-configurable %subdataset macro filter as an additional restriction on the resulting rows.

11. Derive display fields: ODB Visit is the visit mnemonic with 'CCRF' relabeled to 'Common'; ODB Sequence is displayed in the form 'formidx(itemsetidx)'; ODB Stop Date is displayed as 'Continuing' whenever the underlying computed stop date was originally null (i.e., it was defaulted to the system date in step 3) rather than showing the defaulted system date value.

12. Sort the intermediate result (called 'finalini') by subjectid, then produce the final output sorted by Country, Site Number, Subject Number, ODB Visit, ODB Sequence (form-level id), ODB Add-on Sequence (item-set id), AE Sequence, and ODB Form.

13. If, after all the above filtering, zero qualifying records remain, output a single placeholder message stating 'No records meet the test condition' instead of the detail listing; otherwise output the full detail listing with one row per Subject/ODB record/AE combination.

3. Spec Review

Readiness: Ready, with notes — The spec is largely thorough and internally consistent, with clear logic for the core ODB/SAE/CM reconciliation matching, but has several warning-level gaps around exact-match tolerance for CM reconciliation, multi-AE-match handling per ODB record, AE Term/Sequence field sourcing, and a naming mismatch between the Column Heading and Sorting sections for ODB Sequence vs. ODB Add-on Sequence. None of these rise to blocking, but they should be clarified before generation to avoid the query generator making silent assumptions that affect output rows.

WARNING (completeness): The Column Heading list includes 'AE Sequence' and 'AE Term', but the Logic section never explicitly states that AE Sequence or AE Term are pulled from the AE form as display fields -- it only describes AE Onset Date and the SAE/ECI qualification logic. There's no explicit mapping showing which AE field populates 'AE Term' or confirming 'AE Sequence' is the AE form's sequence number. → Add an explicit line under Derived/display fields (or Logic) stating the source field for 'AE Term' (e.g., the verbatim/preferred AE term field) and confirming 'AE Sequence' is the AE record's form sequence number, so the query generator doesn't have to infer it from context.

WARNING (completeness): Step 2 of the matching logic describes overlap exclusion only in terms of a single SAE/ECI window per ODB record, but an ODB record could overlap multiple qualifying SAE/ECI events for the same subject (e.g., two SAEs 20 days apart both within 14 days of a long ODB range). The spec doesn't state whether multiple AE matches per ODB record should each produce a separate output row or be deduplicated. → Clarify explicitly: 'if an ODB record overlaps more than one qualifying SAE/ECI window, output one row per ODB/AE combination' (which the Column Heading/one-row-per-combination language in Scenario 2 suggests, but this should be stated unambiguously in the matching logic steps, not just the scenario description).

WARNING (ambiguity): Step 3's CM match requires 'CM Imputed Start Date equals the ODB record's computed start date' exactly, with no tolerance window. Given the report's stated purpose is to catch under-reporting of oncologic treatments in CM 'relative to ODB source records,' an exact-date-only match could produce false positives (flagging as unreconciled a CM record that is clearly the same drug course but logged with a slightly different start date) -- the spec doesn't state whether this exact-match is intentional or whether a tolerance/window was considered. → State explicitly whether the CM/ODB match must be an exact start-date match (with rationale) or whether some date tolerance is intended, so the generated query's reconciliation logic reflects the actual business intent rather than an assumed strict interpretation.

INFO (ambiguity): The SAE/ECI onset-window is stated as 'onset date minus 14 days to onset date plus 14 days' but doesn't specify whether this is inclusive of the boundary days (e.g., is a start date exactly 14 days before onset included?). The exclusion rule 'exclude when ODB stop date is before (onset-14) or ODB start date is after (onset+14)' does imply inclusivity by using 'before'/'after' rather than '<=' /'>=', but this should be stated plainly. → Add a one-line clarification: 'the 14-day window is inclusive of both boundary dates' to remove any residual doubt for the query generator.

INFO (consistency): The Description says the CM match should have 'no matching record' in CM, while the matching logic in step 3-4 defines 'matching' narrowly as Medication Category='Oncologic' + same standardized name + same exact start date. The Description doesn't mention Medication Category as part of the matching criteria, which could read as if any same-drug/same-date CM record (regardless of category) should count as reconciled. → Update the Description to mention that only CM records categorized as 'Oncologic' are considered as candidate matches, so the description and detailed logic agree on what counts as a 'matching CM record'.

WARNING (naming): The ODB Sequence field is described as displayed as 'formidx(itemsetidx)', but the Sorting section separately lists 'ODB Sequence (form-level id)' and 'ODB Add-on Sequence (item-set id)' as two distinct sort keys. The Column Heading list, however, only shows a single 'ODB Sequence' column with no separate 'ODB Add-on Sequence' heading. → Clarify whether ODB Add-on Sequence is a separate column that should appear in the Column Heading list, or whether it's only used internally for sorting and displayed combined within the single 'ODB Sequence' field as 'formidx(itemsetidx)'. Align the Column Heading and Sorting sections to use the same field names.

INFO (testability): Report Scenarios 3, 4, and 5 describe conditional behaviors (form-filter parameter, older-study fallback, missing-stop-date handling) rather than distinct, mutually exclusive report outputs like Scenarios 1 and 2. It's unclear whether these are meant to be verified as separate test cases against distinct sample datasets or are just narrative notes on logic already covered in the Logic section. → For test planning purposes, restate Scenarios 3-5 as concrete test conditions with expected sample data and expected output rows (e.g., 'given a subject with only

regimen-level dates, verify the report falls back correctly'), rather than as logic restatements, so testers can distinguish them operationally.

WARNING (completeness): The Logic states the DOV error flag excludes 'erroneous visits' for both AE-related visit data and ODB records, but AE records themselves are only filtered by 'non-null onset date and not marked as deleted' -- there's no statement about whether AE records tied to an erroneous visit (per DOV error flag) are also excluded, unlike ODB records which explicitly have this exclusion. → Clarify whether AE records are also subject to DOV-error-flag exclusion the same way ODB records are, or explicitly state that this exclusion only applies to ODB/visit-level data and not to AE records.

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  ODB.Country AS `Country`,
  ODB.SiteNumber AS `Site`,
  ODB.SubjectNumber AS `Subject Number`,
  ODB.VisitMnemonicDisp AS `ODB Visit`,
  ODB.OdbFormName AS `ODB Form`,
  ODB.OdbSeqDisp AS `ODB Sequence`,
  ODB.OdbName AS `ODB Name`,
  DATE_FORMAT(ODB.OdbStartDateRaw, '%d%b%Y') AS `ODB Start Date`,
  CASE WHEN ODB.OdbStopDateOrig IS NULL THEN 'Continuing' ELSE DATE_FORMAT(ODB.OdbStopDateRaw,
'%d%b%Y') END AS `ODB Stop Date`,
  ODB.AeSeq AS `AE Sequence`,
  ODB.AeTerm AS `AE Term`,
  DATE_FORMAT(ODB.AeOnsetDateRaw, '%d%b%Y') AS `AE Onset Date and Time`
FROM
(
  SELECT
    `@HDR`.`Subject`.`ID` AS SubjectID,
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS SiteNumber,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@Form`.`Name` AS OdbFormName,
    CASE WHEN `@HDR`.`Event`.`Name` = 'CCRF' THEN 'Common' ELSE `@HDR`.`Event`.`Name` END AS
VisitMnemonicDisp,
    CONCAT(ODBHFFM.ODBHFCMSPIDIT, '(', ODBHFFM.ODBHFSPIDASSOCIT, ')') AS OdbSeqDisp,
    ODBHFFM.ODBHFCMTRTIT AS OdbName,
    COALESCE(RAWDATE(ODBHFFM.ODBHFCMSTDTCIT), RAWDATE(ODBHFFM.ODBHFCMSTDTIIT)) AS OdbStartDateRaw,
    COALESCE(RAWDATE(ODBHFFM.ODBHFCMENDTCIT), RAWDATE(ODBHFFM.ODBHFCMENDTIIT)) AS OdbStopDateOrig,
    COALESCE(RAWDATE(ODBHFFM.ODBHFCMENDTCIT), RAWDATE(ODBHFFM.ODBHFCMENDTIIT), CURRENT_DATE) AS
OdbStopDateRaw,
    AE.AeSeq AS AeSeq,
    AE.AeTerm AS AeTerm,
    AE.AeOnsetDateRaw AS AeOnsetDateRaw
  FROM `EDC`.`ODBHFFM` ODBHFFM
  INNER JOIN
  (
    SELECT
      `@HDR`.`Subject`.`ID` AS SubjectID,
      AEFM.AESPIDIT AS AeSeq,
      AEFM.AEAETERMIT AS AeTerm,
      RAWDATE(AEFM.AESTDTCIT) AS AeOnsetDateRaw
    FROM `EDC`.`AEFM` AEFM
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (AEFM.DELETEDFM = 'N' OR AEFM.DELETEDFM IS NULL)
      AND (AEFM.DELETEDSN = 'N' OR AEFM.DELETEDSN IS NULL)
      AND AEFM.AESTDTCIT IS NOT NULL
      AND (
        (AEFM.AEAESERCIT IS NOT NULL AND AEFM.AEAESERCNCB <> 'None')
        OR AEFM.AEAECLINTIT = 'Yes'
      )
  ) AE
  ON `@HDR`.`Subject`.`ID` = AE.SubjectID
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ODBHFFM.DELETEDFM = 'N' OR ODBHFFM.DELETEDFM IS NULL)
    AND (ODBHFFM.DELETEDSN = 'N' OR ODBHFFM.DELETEDSN IS NULL)
    AND ODBHFFM.ODBHFCMTRTIT IS NOT NULL
    AND COALESCE(RAWDATE(ODBHFFM.ODBHFCMSTDTCIT), RAWDATE(ODBHFFM.ODBHFCMSTDTIIT)) IS NOT NULL
    AND NOT (
      COALESCE(RAWDATE(ODBHFFM.ODBHFCMENDTCIT), RAWDATE(ODBHFFM.ODBHFCMENDTIIT), CURRENT_DATE) <
SUBDATE(AE.AeOnsetDateRaw, INTERVAL 14 DAY)
      OR COALESCE(RAWDATE(ODBHFFM.ODBHFCMSTDTCIT), RAWDATE(ODBHFFM.ODBHFCMSTDTIIT)) >
SUBDATE(AE.AeOnsetDateRaw, INTERVAL -14 DAY)
    )
) ODB
WHERE NOT EXISTS (
  SELECT 1
  FROM `EDC`.`CMFM` CMFM
  WHERE `@HDR`.`Subject`.`ID` = ODB.SubjectID
    AND `@Form`.`Status` = 'submitted__v'
```

```
    AND (CMFM.DELETEDFM = 'N' OR CMFM.DELETEDFM IS NULL)
    AND LOWER(CMFM.CMSCATIT) = 'oncologic'
    AND LOWER(CMFM.CMTRTIT) = LOWER(ODB.OdbName)
    AND RAWDATE(CMFM.CMSTDTCIT) = ODB.OdbStartDateRaw
)
ORDER BY
  `Country`, `Site`, `Subject Number`, `ODB Visit`, `ODB Sequence`, `AE Sequence`, `ODB Form`
```

I modeled this as: an ODB derived table (using ODBHFFM as the representative ODB add-on form variant per the "Known Forms" list, since ODB/ODBF/ODBCF/ODBCS/ODBH/ODBNF variants weren't individually present in metadata) inner-joined to a qualifying SAE/ECI AE subquery on Subject, filtered to the 14-day overlap window, then a correlated NOT EXISTS against CMFM (using CMSCATIT as Medication Category, CMTRTIT as Standardized Medication Name, CMSTDTCIT as Imputed Start Date) to exclude reconciled records. Key assumptions flagged: only ODBHFFM was available in the known-items list to represent the unified "ODB dataset" (other physical ODB form variants like ODB, ODBF, ODBCF, ODBCS, ODBH, ODBNF are not in the provided metadata, so this query only covers ODBHFFM data — please confirm additional Form names via DESCRIBE FORM and union them in if present); SAE flag logic uses AEAESERCIT (Serious Criteria composite) non-null AND AEAESERCNCB (Is this a serious AE) <> 'None', with AEAECLINTIT as the Event of Clinical Interest flag; ODB Sequence is displayed as "formidx(itemsetidx)" using ODBHFCMSPIDIT and ODBHFSPIDASSOCIT; the %odbform/%subdataset macro-driven optional filters and zero-result placeholder message are report-tool-level behaviors not expressible as static CQL and were omitted, and UNK partial-date imputation was not implemented since dates were assumed pre-parsed date items rather than raw DTC strings containing "UNK".